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CE F231 Fluid Mechanics

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Lecture 35: Dimensional and Model Analyses

Learning Outcome

- Determine relation between dependent and independent variables in a physical phenomenon
 - Rayleigh's method
 - Buckingham's π method
- Predict performance/characteristics of prototype based on studies/experiments performed on models

Dimensions

<i>S. No.</i>	<i>Physical Quantity</i>	<i>Symbol</i>	<i>Dimensions</i>
	(a) Fundamental		
1.	Length	L	L
2.	Mass	M	M
3.	Time	T	T
	(b) Geometric		
4.	Area	A	L^2
5.	Volume	$\forall$	L^3
	(c) Kinematic Quantities		
6.	Velocity	v	LT^{-1}
7.	Angular Velocity	ω	T^{-1}
8.	Acceleration	a	LT^{-2}
9.	Angular Acceleration	α	T^{-2}
10.	Discharge	Q	L^3T^{-1}
11.	Acceleration due to Gravity	g	LT^{-2}
12.	Kinematic Viscosity	ν	L^2T^{-1}

Dimensions

(d) Dynamic Quantities

13.	Force	F	MLT^{-2}
14.	Weight	W	MLT^{-2}
15.	Density	ρ	ML^{-3}
16.	Specific Weight	w	$ML^{-2}T^{-2}$
17.	Dynamic Viscosity	μ	$ML^{-1}T^{-1}$
18.	Pressure Intensity	p	$ML^{-1}T^{-2}$
19.	Modulus of Elasticity	$\left\{ \frac{K}{E} \right\}$	$ML^{-1}T^{-2}$
20.	Surface Tension	σ	MT^{-2}
21.	Shear Stress	τ	$ML^{-1}T^{-2}$
22.	Work, Energy	W or E	ML^2T^{-2}
23.	Power	P	ML^2T^{-3}
24.	Torque	T	ML^2T^{-2}
25.	Momentum	M	MLT^{-1}

Dimensional homogeneity

- An equation representing a physical phenomenon must be dimensionally homogeneous
- Dimensionally homogeneous → powers of fundamental dimensions on LHS and RHS must be same

$$Q = \frac{2}{3} C_d \sqrt{2g} LH^{3/2} \quad \dots \text{(Flow over rectangular weir)}$$

$$Q = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} H^{5/2} \quad \dots \text{(Flow over triangular weir)}$$

$$\Delta p = \frac{32\mu VL}{D^2} \quad \dots \text{(Laminar flow in pipes)}$$

$$\Delta p = f \frac{l}{D} \frac{\rho V^2}{2} \quad \dots \text{(Turbulent flow in pipes)}$$

Methods of dimensional analysis

- If the variables in a physical phenomenon are known, relation among dependent and independent variables can be obtained using methods of dimensional analysis
 - Rayleigh's method → max. 3 to 4 variables
 - Difficult if number of variables > number of fundamental dimensions involved in the physical phenomenon
 - Buckingham's π → overcomes difficulties in Rayleigh's method

Rayleigh's method

- If Y is a variable that depends on X_1 , X_2 , and X_3
- $Y = f(X_1, X_2, X_3)$
- $Y = k X_1^a X_2^b X_3^c$
- k is a constant and a , b , and c are arbitrary powers
- Determine a , b , and c satisfying dimensional homogeneity

Rayleigh's method example

Power P developed by a pump depends on head H , discharge Q , and unit weight γ . Find expression for power

$$P = ML^2 T^{-3}; H = L; Q = L^3 T^{-1}; \gamma = ML^{-2} T^{-2}$$

➤ $P = f(H, Q, \gamma)$ $ML^2 T^{-3} = L^a \times (L^3 T^{-1})^b \times (ML^{-2} T^{-2})^c$

$$1 = c$$

➤ $P = k H^a Q^b \gamma^c$ $2 = a + 3b - 2c = a + 3b - 2 \Rightarrow a = 1$

$$-3 = -b - 2c \Rightarrow -3 = -b - 2 \Rightarrow b = 1$$

➤ k is a constant and a , b , and c are arbitrary powers

$$P = k H^1 Q^1 \gamma^1$$

➤ Determine a , b , and c satisfying dimensional homogeneity

Buckingham's π method

The variables controlling the motion of a floating vessel through water are the drag force F , the speed V , the length L , the density ρ and dynamic viscosity μ of water and acceleration due to gravity g . Derive an expression for F by dimensional analysis.

$$\left[\text{Ans. } F = \rho L^2 V^2 \phi \left[\frac{\mu}{\rho V L}, \frac{Lg}{V^2} \right] \right]$$

$$F = f(V, L, \rho, \mu, g)$$

If there are n variables (independent + dependent) in a physical phenomenon and if these variables contain m fundamental dimensions, then the variables are arranged into $(n-m)$ dimensionless terms. Each term is called a π term.

$$n = 6$$

$$m = 3$$

$$n - m = 3$$

ie 3 π terms,

π_1, π_2 , and π_3

$$F = M L T^{-2}$$

$$V = L T^{-1}$$

$$L = L$$

$$\rho = M L^{-3}$$

$$\mu = M L^{-1} T^{-1}$$

$$g = L T^{-2}$$

$m = 3$

$M,$

L and

T

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Each π term contains $m+1$ variables, where m is the number of fundamental dimensions, and these m variables are called repeating variables.

$m+1 = 4$

$\pi_1 =$	_____	_____	_____	_____
$\pi_2 =$	_____	_____	_____	_____
$\pi_3 =$	_____	_____	_____	_____

→ To be filled by repeating variables

Buckingham's π method

12.4.3 Method of Selecting Repeating Variables. The number of repeating variables are equal to the number of fundamental dimensions of the problem. The choice of repeating variables is governed by the following considerations :

1. As far as possible, the dependent variable should not be selected as repeating variable.

$\pi_1 =$
 $\pi_2 =$
 $\pi_3 =$

→ should not be filled by the dependent variable, ie F in this example

Buckingham's π method

12.4.3 Method of Selecting Repeating Variables. The number of repeating variables are equal to the number of fundamental dimensions of the problem. The choice of repeating variables is governed by the following considerations :

1. As far as possible, the dependent variable should not be selected as repeating variable.
2. The repeating variables should be chosen in such a way that one variable contains geometric property, other variable contains flow property and third variable contains fluid property.

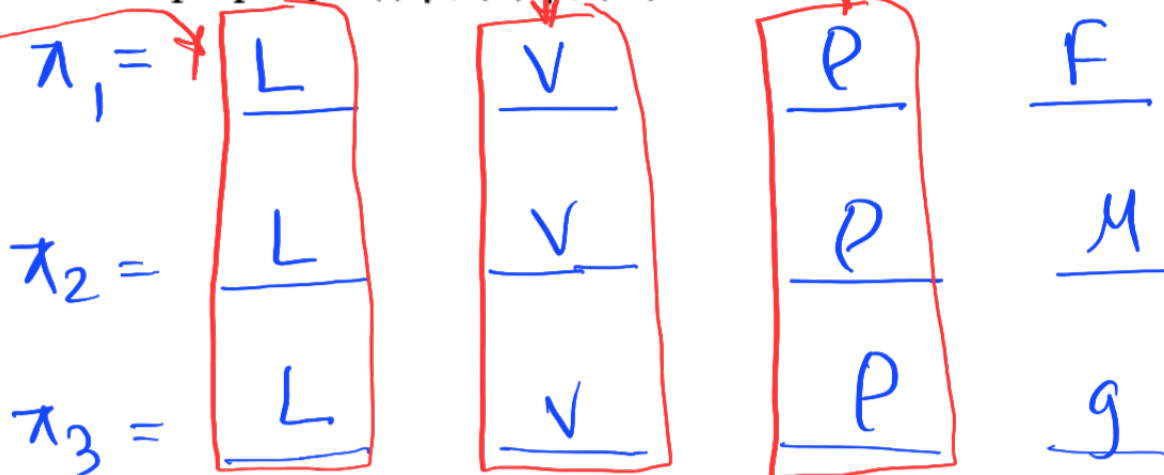
Variables with Geometric Property are

- (i) Length, l (ii) d (iii) Height, H etc.

Variables with flow property are

- (i) Velocity, V (ii) Acceleration etc.

Variables with fluid property : (i) μ , (ii) ρ , (iii) ω etc.



$$\begin{array}{l} \pi_1 = \frac{L}{L} \quad \frac{V}{V} \quad \frac{\rho}{\rho} \quad \frac{F}{F} \\ \pi_2 = \frac{L}{L} \quad \frac{V}{V} \quad \frac{\rho}{\rho} \quad \frac{\mu}{\mu} \\ \pi_3 = \frac{L}{L} \quad \frac{V}{V} \quad \frac{\rho}{\rho} \quad \frac{g}{g} \end{array}$$

Buckingham's π method

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1. As far as possible, the dependent variable should not be selected as repeating variable.
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Variables with flow property are

- (i) Velocity, V (ii) Acceleration etc.

Variables with fluid property : (i) μ , (ii) ρ , (iii) ω etc.

3. The repeating variables selected should not form a dimensionless group.

$$\begin{aligned} \hookrightarrow L \times V \times \rho &= L \times L T^{-1} \times M L^{-3} \\ &= M L^{-1} T^{-1} \rightarrow \text{not a dimensionless group} \end{aligned}$$

Buckingham's π method

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- (i) Length, l (ii) d (iii) Height, H etc.

Variables with flow property are

- (i) Velocity, V (ii) Acceleration etc.

Variables with fluid property : (i) μ , (ii) ρ , (iii) ω etc.

3. The repeating variables selected should not form a dimensionless group.
4. The repeating variables together must have the same number of fundamental dimensions.

↪ L, V, ρ together have same number of fundamental dimensions, i.e. 3.

Buckingham's π method

12.4.3 Method of Selecting Repeating Variables. The number of repeating variables are equal to the number of fundamental dimensions of the problem. The choice of repeating variables is governed by the following considerations :

1. As far as possible, the dependent variable should not be selected as repeating variable.
2. The repeating variables should be chosen in such a way that one variable contains geometric property, other variable contains flow property and third variable contains fluid property.

Variables with Geometric Property are

- | | | |
|-----------------|----------|------------------------|
| (i) Length, l | (ii) d | (iii) Height, H etc. |
|-----------------|----------|------------------------|

Variables with flow property are

- | | |
|-------------------|------------------------|
| (i) Velocity, V | (ii) Acceleration etc. |
|-------------------|------------------------|

Variables with fluid property : (i) μ , (ii) ρ , (iii) ω etc.

3. The repeating variables selected should not form a dimensionless group.
4. The repeating variables together must have the same number of fundamental dimensions.
5. No two repeating variables should have the same dimensions.

 L, V, ρ don't have the same dimensions

Buckingham's π method

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Variables with Geometric Property are

- | | | |
|-----------------|----------|------------------------|
| (i) Length, l | (ii) d | (iii) Height, H etc. |
|-----------------|----------|------------------------|

Variables with flow property are

- | | |
|-------------------|------------------------|
| (i) Velocity, V | (ii) Acceleration etc. |
|-------------------|------------------------|

Variables with fluid property : (i) μ , (ii) ρ , (iii) ω etc.

3. The repeating variables selected should not form a dimensionless group.
4. The repeating variables together must have the same number of fundamental dimensions.
5. No two repeating variables should have the same dimensions.

In most of fluid mechanics problems, the choice of repeating variables may be (i) d, v, ρ (ii) l, v, ρ or (iii) l, v, μ or (iv) d, v, μ .

Buckingham's π method

The variables controlling the motion of a floating vessel through water are the drag force F , the speed V , the length L , the density ρ and dynamic viscosity μ of water and acceleration due to gravity g . Derive an expression for F by dimensional analysis.

$$\left[\text{Ans. } F = \rho L^2 V^2 \phi \left[\frac{\mu}{\rho V L}, \frac{Lg}{V^2} \right] \right]$$

$$\pi_1 = \frac{L^{a_1}}{\phantom{L^{a_1}}} \times \frac{V^{b_1}}{\phantom{V^{b_1}}} \times \frac{\rho^{c_1}}{\phantom{\rho^{c_1}}} \times \frac{F}{}$$

$$\pi_2 = \frac{L^{a_2}}{\phantom{L^{a_2}}} \times \frac{V^{b_2}}{\phantom{V^{b_2}}} \times \frac{\rho^{c_2}}{\phantom{\rho^{c_2}}} \times \frac{\mu}{}$$

$$\pi_3 = \frac{L^{a_3}}{\phantom{L^{a_3}}} \times \frac{V^{b_3}}{\phantom{V^{b_3}}} \times \frac{\rho^{c_3}}{\phantom{\rho^{c_3}}} \times \frac{g}{}$$

Buckingham's π method

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$$\left[\text{Ans. } F = \rho L^2 V^2 \phi \left[\frac{\mu}{\rho V L}, \frac{Lg}{V^2} \right] \right]$$

for π_1

$$M^0 L^0 T^0 = L^{a_1} \times (L T^{-1})^{b_1} \times (M L^{-3})^{c_1} \times (M L T^{-2})^{d_1}$$

$$\Rightarrow 0 = c_1 + 1 \Rightarrow c_1 = -1$$

$$0 = a_1 + b_1 - 3c_1 + 1 \Rightarrow 0 = a_1 - 2 + 3 + 1 \Rightarrow a_1 = -2$$

$$0 = -b_1 - 2 \Rightarrow b_1 = -2$$

$$\therefore \pi_1 = L^{-2} V^{-2} \times \rho^{-1} \times F \Rightarrow \pi_1 = \frac{F}{\rho L^2 V^2}$$

Buckingham's π method

The variables controlling the motion of a floating vessel through water are the drag force F , the speed V , the length L , the density ρ and dynamic viscosity μ of water and acceleration due to gravity g . Derive an expression for F by dimensional analysis.

$$\left[\text{Ans. } F = \rho L^2 V^2 \phi \left[\frac{\mu}{\rho V L}, \frac{Lg}{V^2} \right] \right]$$

For π_2

$$M^0 L^0 T^0 = L^{a_2} \times (L T^{-1})^{b_2} \times (M L^{-3})^{c_2} \times (M L^{-1} T^{-1})^{d_2}$$

$$\Rightarrow 0 = c_2 + 1 \Rightarrow c_2 = -1$$

$$0 = a_2 + b_2 - 3c_2 - 1 \Rightarrow 0 = a_2 - 1 + 3 - 1 \Rightarrow a_2 = -1$$

$$0 = -b_2 - 1 \Rightarrow b_2 = -1$$

$$\therefore \pi_2 = L^{-1} \times V^{-1} \times \rho^{-1} \times \mu \Rightarrow \pi_2 = \frac{\mu}{\rho V L}$$

Buckingham's π method

The variables controlling the motion of a floating vessel through water are the drag force F , the speed V , the length L , the density ρ and dynamic viscosity μ of water and acceleration due to gravity g . Derive an expression for F by dimensional analysis.

$$\left[\text{Ans. } F = \rho L^2 V^2 \phi \left[\frac{\mu}{\rho V L}, \frac{Lg}{V^2} \right] \right]$$

for π_3

$$M^0 L^0 T^0 = L^{a_3} \times (L T^{-1})^{b_3} \times (M L^{-3})^{c_3} \times (L T^{-2})^1$$

$$0 = c_3$$

$$0 = a_3 + b_3 - 3c_3 + 1 \Rightarrow 0 = a_3 - 2 - 0 + 1 \Rightarrow a_3 = 1$$

$$0 = -b_3 - 2 \Rightarrow b_3 = -2$$

$$\therefore \pi_3 = L^1 V^{-2} \rho^0 g^1 \Rightarrow \pi_3 = \frac{Lg}{V^2}$$

Buckingham's π method

The variables controlling the motion of a floating vessel through water are the drag force F , the speed V , the length L , the density ρ and dynamic viscosity μ of water and acceleration due to gravity g . Derive an expression for F by dimensional analysis.

$$\left[\text{Ans. } F = \rho L^2 V^2 \phi \left[\frac{\mu}{\rho V L}, \frac{Lg}{V^2} \right] \right]$$

As per Buckingham's π theorem

$$f(\pi_1, \pi_2, \pi_3) = 0 \rightarrow \text{some function}$$

$$\Rightarrow \pi_1 = \phi(\pi_2, \pi_3) \rightarrow \text{some other function}$$

$$\Rightarrow \frac{F}{\rho L^2 V^2} = \phi \left(\frac{\mu}{\rho V L}, \frac{Lg}{V^2} \right)$$

$$\Rightarrow F = \rho L^2 V^2 \phi \left(\frac{\mu}{\rho V L}, \frac{Lg}{V^2} \right)$$

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THANK YOU